PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 8 · Data collection and cleansing

TUESDAY OCT 13 AND THURSDAY OCT 15, 2026

SETUP

UPDATE BOTH FOLDERS BEFORE WE START

1 Open Claude Code in your repo folder, spax402-your-username.

Your personal repository sits next to spax402-course-materials, the shared folder Professor Davis updates weekly.

2 Run one prompt.

The /update skill brings this week's files into your personal repo without saving over your work, and the course materials folder updates by asking Claude to pull the latest version.

> Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials. ↵

3 Confirm that today's deck is in spax402-course-materials/decks/.

Thursday's files arrive through /update before Thursday's class.

01

DATA BEFORE MODELS

OFTEN, THE QUALITY AND STRUCTURE OF DATA THAT IS ANALYZED IS MORE IMPORTANT THAN THE MODEL.



WHAT INPUTS WOULD YOU CONSIDER WHEN PREDICTING WHIFF PROBABILITY ON A CHANGEUP?

WHY GET DOMAIN EXPERT INPUT?

IDENTIFY PRIORITIES

Coaches know which questions matter this season.

IMPROVE THE PRODUCT

The person who watches every game spots the column that is measured wrong.

CREATE BUY-IN

Staff use what they helped build.

02

WHAT ONE ROW IS

KNOW WHAT ONE ROW OF A TABLE REPRESENTS.

One pick in a draft. One player in one season. One team in one game.

Counts, averages and joins all depend on it.

ONE ROW

2025 NFL DRAFT, FIRST ROUND

ROUND	PICK	TEAM	PLAYER	POS	COLLEGE
1	1	TEN	Cam Ward	QB	Miami (FL)
1	2	JAX	Travis Hunter	WR	Colorado
1	3	NYG	Abdul Carter	DE	Penn St.
1	4	NWE	Will Campbell	OT	LSU
1	5	CLE	Mason Graham	DT	Michigan
1	6	LVR	Ashton Jeanty	RB	Boise St.
1	7	NYJ	Armand Membou	OL	Missouri
1	8	CAR	Tetairoa McMillan	WR	Arizona

Source: nflverse draft_picks release file.

ASK CLAUDE

1 The prompt.

> What was the average 2025 draft pick of players from Ohio State and from Miami? ↵

ONE ROW

AVERAGE 2025 DRAFT PICK

SCHOOL	PICKS	AVERAGE PICK
Ohio St.	14	19
Miami (FL)	7	16

? WAS OHIO STATE'S AVERAGE 2025 PICK REALLY 19?

ONE ROW

2025 NFL DRAFT BY PICK AND BY SCHOOL

ONE ROW = ONE PICK

Pick	Player	College

and 249 more picks

257 rows

ONE ROW = ONE SCHOOL IN ONE DRAFT

College	Draft	Picks
Ohio St.	2025	**14**
Miami (FL)	2025	**7**

and 84 more schools

86 rows

Ask Claude for picks by school and you could get either table. Say what one row should be.

Source: nflverse draft_picks release file.

ONE ROW

2023-24 NBA SEASON TOTALS

PLAYER	GAMES	POINTS
Jayson Tatum	74	1,987
Domantas Sabonis	82	1,593
Pascal Siakam	41	872



WHAT DOES ONE ROW OF THIS TABLE REPRESENT?

? CLAUDE SUMS GAMES BY PLAYER. WHAT DOES IT REPORT FOR SIAKAM IN 2023-24?

CHECK FOR REPEATED ROWS

1 Before you sum.

I believe each row of this table is one player's 2023-24 season. Is that true? If not, show me the rows that break it and tell me what they are. ↵

03

UNIQUE IDS AND JOINS

PLAYER IDS IN THE DRAFT FILE, 2015 TO 2025

COLUMN	ISSUED BY	MISSING
pfr_player_id	Pro Football Reference	0.0%
gsis_id	The NFL	0.3%
cfb_player_id	Sports Reference, college	6.3%

Join on an ID when both tables have one.

Source: nflverse draft_picks release file.



YOU JOIN DRAFT PICKS TO COLLEGEFOOTBALLDATA'S TEAM LIST BY SCHOOL NAME. WHAT COULD GO WRONG?

CROSSWALK, DRAFT NAME TO COLLEGEFOOTBALLDATA NAME

SCHOOL ID	DRAFT FILE	COLLEGEFOOTBALLDATA
ohio-state	Ohio St.	Ohio State
miami-fl	Miami (FL)	Miami
miami-oh	Miami (OH)	Miami (OH)
delaware	Delaware	Delaware
delaware-state	Delaware St.	Delaware State

One ID per school, with each source's spelling. Nothing is guessed at join time.

HAVE CLAUDE DRAFT THE CROSSWALK

Claude writes the crosswalk once. The join reads it every run.

> List every school in the draft file that has no exact match in the CollegeFootballData team list. Suggest a match for each, flag any you're unsure of, and save the result as crosswalk.csv. Then join through it. ↵

THURSDAY'S TWO TABLES

draft_picks	
season	integer
round	integer
pick	integer
team	text
pfr_player_id	text
pfr_player_name	text
position	text
college	text
w_av	number
and 27 more columns	

?

recruiting/teams	
year	integer
rank	integer
team	text
points	number

A schema lists a table's columns and what type each one holds. The draft file has 36 columns; nine are shown.

Sources: nflverse draft_picks; CollegeFootballData API documentation, /recruiting/teams.



WHICH COLUMNS IDENTIFY ONE ROW IN EACH TABLE? WHICH COLUMNS JOIN THEM?

DRAFT YEAR TO FINAL COLLEGE SEASON

DRAFT	FOOTBALL SEASON	BASKETBALL SEASON
2024	2023	2023-24
2025	2024	2024-25
2026	2025	2025-26

The 2026 draft follows the 2025 football season and the 2025-26 basketball season. Join on the converted year, not the label.

04

BEFORE YOU USE A DATA SOURCE

FOUR STEPS BEFORE YOU USE A DATA SOURCE

1 **Profile.**

Rows and columns, what one row is, which columns identify a row, and the share of blanks in every column.

2 **Look.**

Plot counts over time and the distribution of each column that matters, and read a handful of raw rows.

3 **Check.**

Ask how many rows were read, kept and dropped, and compare a total you already know.

4 **Record.**

Where and when the data came from, and every change you made to it.

PROFILE THE DRAFT FILE

1 Load it.

Download the nflverse draft picks file from https://github.com/nflverse/nflverse-data/releases/download/draft_picks/draft_picks.csv and keep the 2015 to 2025 drafts. How many rows and columns? What is one row? Which columns could identify a row, and are any duplicated?

2 Count the blanks.

For every column, what share of values is blank? Sort from most blank to least.



WHICH BLANKS ARE A PROBLEM, AND WHICH WOULD YOU EXPECT?

LOOK AT THE DRAFT FILE

Plot it.

> Plot picks by position for each draft from 2015 to 2025, and plot age at the draft. Show me ten random rows. Tell me anything that looks wrong. ↵

? PICKS LABELED T, FOR OFFENSIVE TACKLE, FELL FROM 22 IN 2020 TO 0 IN 2024. WHAT HAPPENED?

CHECK THE DRAFT FILE

Rows read, kept and dropped.

How many rows did you read, keep and drop for each draft from 2015 to 2025, and why was each dropped row dropped? Every draft in those years had between 253 and 262 picks. Flag any year outside that range.

CLEAN THE POSITION LABELS

Group the labels.

Add a column called position_group that maps T, OT, G, C and OL to OL. Keep the original position column. Count offensive line picks per draft before and after, and confirm the total picks per draft didn't change.

RECORD THE DRAFT FILE

Before you change anything.

Save the raw file as downloaded and don't edit it. Write a short README next to it: the source URL, today's date, the filters you applied, rows before and after, and every change you made, with the reason for each.



STATCAST HAD NO EXIT VELOCITY FOR SOME BATTED BALLS IN 2015. WHICH ONES DO YOU THINK WENT MISSING?

LOOK

SIX PLAYS, SUCCESS AS COMPUTED

DOWN	PLAY	GAIN	NEEDED	SUCCESS
1st & 10	Pass, complete	7	4	Yes
2nd & 3	Run	1	2	No
1st & 10	Pass, defensive pass interference	0, plus 12 penalty	4	No
3rd & 6	Pass, complete	9	6	Yes
1st & 10	Run	2	4	No
2nd & 8	Sack	-7	5	No

Success rate says 2 of 6. Which row would you question?

Illustration. A play is a success when it gains 40% of the yards to go on 1st down, 60% on 2nd, and all of them on 3rd or 4th.

05

WHERE DATA COMES FROM

WHERE TO GET DATA

1 **A published download.**

nflverse, Baseball Savant, Kaggle, FiveThirtyEight on GitHub, MoneyPuck. One file, the same for everyone.

2 **An API.**

CollegeFootballData, the MLB Stats API, the NHL API. A supported way for a program to ask for data.

3 **Scraping, last.**

A program reads pages built for people. Check the site's terms and robots.txt first.

PARTS OF AN API CALL

PART	EXAMPLE	THINK OF IT AS
Base URL	api.collegefootballdata.com	The database
Endpoint	/recruiting/teams	One table in it
Parameters	?year=2020	A filter on that table
Key	C5wLFxbTXs4k0u9OF8PDqRtuKSRZ2G	Your ID card
Response	wr Rows of year, rank, team, points	The rows that matched
Status code	200, 401, 429	Worked, bad key, slow down
Limit	1,000 calls a month, free	How many times you can ask

An endpoint is a table you can ask for. Parameters filter it, and the key says who's asking.

ONE API CALL

THE ADDRESS

```
https://api.collegefootballdata.com/recruiting/teams?year=2020
```

Base URL	Endpoint	Parameter
the database	one table in it	a filter on that table

SENT WITH IT, IN A HEADER

```
Authorization: Bearer C5wLFxbTXs4k0u9OF8PDqRtuKSRZ2Gwr
```

CollegeFootballData takes the key here, outside the address.

SOME APIS TAKE IT IN THE ADDRESS INSTEAD

```
...?year=2020&api_key=C5wLFxbTXs4k0u9OF8PDqRtuKSRZ2Gwr
```

Then the key is saved wherever the address is: browser history, server logs, a pasted link.

The base URL, endpoint and parameters are one web address. The key travels beside it, not in it.

CollegeFootballData API documentation.

HAVE CLAUDE READ THE API DOCS BEFORE IT WRITES CODE.

Without the docs it guesses endpoint and field names, and the guesses look right.

AN API KEY IS A PASSWORD.

.ENV

Holds the key on one line:
CFBD_API_KEY=...

.GITIGNORE

Lists .env before the first commit, so the key never reaches GitHub.

.ENV.EXAMPLE

Shows the variable's name with no value, so a teammate knows what to set.

Have your code read the key from a .env file. Keys should be kept out of your chat with an LLM and your repo.



YOU PUSHED A KEY TO A PUBLIC REPO, THEN DELETED THE FILE AND PUSHED AGAIN. ARE YOU SAFE?

06

SCRAPING AND TERMS



CLAUDE WRITES A LOOP THAT PULLS EVERY DIVISION I TEAM PAGE FROM SPORTS REFERENCE. WHAT HAPPENS?

READ THE TERMS BEFORE YOU COLLECT.

The MLB Stats API puts its terms in every response it sends.

Sports Reference has no API because its data licenses do not allow bulk downloads.

TERMS

FBREF ADVANCED STATS, JANUARY 2026

FBREF.COM, PLAYER STANDARD STATS

January 2026

Player	Squad	MP	Min	Gls	Ast

Opta columns removed

Opta's owner ended FBref's feed and asked for the data to be deleted. Licensed data can be taken back.

Source: FBref announcement after Stats Perform (Opta) ended its data agreement.

07

PRIVACY AND OWNERSHIP

NBA PLAYERS CONTROL HOW TEAMS USE THEIR WEARABLE DATA.

The 2017 NBA collective bargaining agreement lets players decline wearables and bars teams from using wearable data in contract or trade talks, with fines up to $250,000.



WHICH LEVEL IS A STUDENT-ATHLETE'S GPA, PAIRED WITH THEIR NAME?

08

THIS WEEK'S WORK

SETUP

UPDATE BOTH FOLDERS BEFORE WE START

1 Open Claude Code in your repo folder, spax402-your-username.

2 Run one prompt.

Thursday's lab files arrive the same way as Tuesday's.

> Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials. ↵

3 Confirm that weeks/week08/README.md is in your repo.

SET UP COLLEGEFOOTBALLDATA

1. Request a key at collegefootballdata.com/key. It arrives by email.
2. Give Claude the documentation or the schema.

> Read the docs at https://api.collegefootballdata.com. Which endpoint returns team recruiting rankings, with what fields, and how does it take the key? (Or use the docs page's download schema option and choose Use in Claude.)

3. Set up .env.

> Using the documentation, set up .env.example with a placeholder for the key, and a .env with the same line. Check .gitignore lists .env.

4. Type your key into .env yourself. Never paste it into the chat.
5. Make one call.

> Read CFBD_API_KEY without printing it. Pull 2020 team recruiting rankings. Show five rows.

WHICH PROGRAMS "DO THE BEST" TURNING RECRUITS INTO DRAFT PICKS?

1 Pull team recruiting rankings for the 2015 to 2020 classes, six calls.

> Profile this pull: rows, columns, what one row is, and the share of blanks in each column. Then plot the number of teams per class year. ↵

2 Download nflverse draft picks for 2018 to 2025, a published file with no key.

> Profile the draft file the same way. Then plot picks per draft year and compare each year with the official count, 253 to 262. ↵

3 Determine what one row is in each, then build a crosswalk to join them.

> How many rows did you read, keep and drop at every filter and join? List the schools that still do not match. Save a note on where each file came from and what you changed. ↵



SOME SCHOOLS DID NOT JOIN. WHY? DO YOU NEED TO CORRECT ANYTHING, OR IS IT OKAY AS IS?



ARE ANY RECRUITING CLASSES TOO EARLY TO CALL?



WHO IS AT THE TOP OF YOUR LEADERBOARD? HOW DID YOU DEFINE "BEST"?

CASE STUDY

WHICH PROGRAMS PRODUCE NFL PLAYERS?

STEPS

1. Finish Thursday's leaderboard.
2. Compare draft slot with career value (w_av), by program and by type of program: conference, Power 4 or Group of 5.
3. Build one interactive view of the comparison.

BRIEF QUESTIONS

1. Which programs are the best at developing recruits into draft picks? What else would you check before finalizing this list?
2. Do NFL teams over- or undervalue players from certain programs or certain types of programs?
3. What other data or analysis would you want to complete before telling a scouting department to adjust its board by school?

Written to an NFL director of college scouting.

CASE STUDY

SCOUTING THE DATA FOR YOUR PROJECT

STEPS

1. List two or three questions for your final project, each with the person who would act on the answer.
2. For each, search for data sources you could use.
3. Pick the question with the best publicly available data for a robust project.

BRIEF QUESTIONS

4. What two or three questions did you consider for your final project? Who would your audience be?
5. For the question you picked, what data sources are available? How will you receive it? (download, API, scrape).

Written to the stakeholder for your final project.

BEFORE YOU PUSH ON MONDAY, OCTOBER 19

Commit Thursday's code and leaderboard, and your career value comparison.

Answer every question in BRIEF.md, in your own words.

Run /audit, then /quiz-me.

Upload your repository's link to Canvas by Monday, October 19 at 11:59pm.